

Rahul Parthy

Big Data Engineer/ ETL Developer / Cloud Engineer

Profile Summary:

Seasoned Big Data Engineer with over **10+** years of hands-on experience in Big Data pipeline architecture (on-premises and cloud) and **11+** years of total IT experience.

Core Expertise:

- **ETL & Data Processing:** Designed and optimized **large-scale ETL/ELT pipelines** using **Spark, PySpark, AWS Glue, and Dataflow (Beam)**, integrating metadata-driven orchestration across **Airflow, Cloud Composer, ADF, and dbt** for robust, scalable data delivery.
- **Real-Time Architectures:** Engineered **event-driven pipelines** leveraging **Kafka, Kinesis, Event Hubs, Pub/Sub, Flink, Spark Streaming, and Beam** for sub-minute analytics and operational alerts.
- **Lakehouse & Data Governance:** Built **compliance-ready data lakehouses** using **S3 (Iceberg/Delta), BigLake/BigQuery, Snowflake, Synapse**, with integrated governance via **Glue, Lake Formation, and Dataplex**.
- **Cross-Cloud ETL/ELT:** Developed **metadata-driven pipelines** across **Airflow, Cloud Composer, Dataflow (Beam), ADF, Glue, dbt, and Snowpipe**, ensuring resilient and scalable data movement.
- **Operational AI Integration:** Combined **streaming and serving layers** through **API Gateway/Apigee, Lambda/Cloud Functions**, and low-latency data stores (**DynamoDB, Redis, HBase**) for responsive AI systems.
- **Hadoop Modernization:** Migrated **YARN/HDFS/Hive/Impala/Presto** ecosystems to cloud lakehouses, reducing costs, improving lineage, and accelerating time-to-insight.

DevOps & Cloud-Native Delivery

- Automated infrastructure provisioning with **Terraform, CloudFormation, Deployment Manager**, and CI/CD via **CodePipeline, GitHub Actions, Azure DevOps, Cloud Build**.
- Containerized and orchestrated workloads using **Docker and Kubernetes**, embedding **data quality gates** and canary release strategies.
- Established **cost, performance, and SLO guardrails** with autoscaling, workload-based scheduling, and observability across **CloudWatch, Azure Monitor, and Cloud Monitoring**.

Governance, Security & Observability

- Enforced **enterprise-grade security** with **IAM/RBAC, KMS/Key Vault/Secret Manager, VPC-SC, and tag-based access control**.
- Implemented **data cataloging, lineage, and quality frameworks** using **Glue Catalog, Purview, Dataplex, Atlas, dbt tests, Great Expectations, and Deequ**.
- Delivered **end-to-end observability** via **CloudWatch, CloudTrail, Grafana, OpenSearch, Azure Monitor, Application Insights, and Cloud Logging**.

Collaboration & Leadership

- Led cross-functional **Agile programs** integrating data engineering, platform, and security to drive measurable outcomes in fraud, treasury, clinical, and grid operations.
- **Mentored engineering teams** in Spark, Scala, Python, streaming architecture, property-based testing, and data contract design.
- Empowered **self-service analytics** through **QuickSight, Power BI, Looker, and Tableau**, backed by governed semantic layers and curated marts.

Technical Skills:

The Architecture of Big Data Systems: Amazon AWS - EC2, S3, Kinesis, Azure, Cloudera Hadoop, Hortonworks Hadoop, Spark, Spark Streaming, Hive, Kafka, Snowflake, Databricks

Programming Languages: Scala, Python, Java, Bash, PySpark

Hadoop Components: Hive, Pig, Zookeeper, Sqoop, Oozie, Yarn, Maven, Flume, HDFS, Airflow

Hadoop Administration: Zookeeper, Oozie, Cloudera Manager, Ambari, Yarn

Data Management: Snowflake, Apache Cassandra, AWS Redshift, Amazon RDS, Apache HBase, SQL, NoSQL, Elasticsearch, HDFS, Data Lake, Data Warehouse, Database, Teradata, SQL Server

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The architecture of ETL Data Pipelines: Apache Airflow, Hive, Sqoop, Flume, Scala, Python, Flume, Apache Kafka, Logstash

Scripting: HiveQL, SQL, Shell Script Language

Big Data Frameworks: Spark/PySpark and Kafka

Spark Framework: Spark API, Spark Streaming, Spark SQL, Spark Structured Streaming

Visualization: Tableau, QlikView, PowerBI

Software Development IDE: Jupyter Notebooks, PyCharm, IntelliJ Continuous Integration (CI-CD)

Versioning: Git, GitHub, Bitbucket

Method: Agile Scrum, Test-Driven Development, Continuous Integration, Unit Testing, Functional Testing, Scenario Testing

Professional Experience

Sr Data/ Cloud Engineer (AWS)

Target Corporation, Minneapolis, MN, Nov'23 – Present

Summary: Led the modernization of legacy data systems into a cloud-native platform on AWS, designed petabyte-scale data pipelines, implemented real-time analytics for retail and supply chain operations, and enabled advanced business intelligence across merchandising, inventory, and customer insights. Responsible for team leadership, architecture design, CI/CD implementation, and end-to-end data governance.

- Designed and implemented a **cloud-native data lakehouse** on AWS using **S3, Glue, Redshift, Lake Formation, Athena, EMR, Kinesis, and Lambda**, consolidating petabyte-scale retail and transactional data, improving query performance by **40%** and reducing storage costs by **25%**.
- Built **real-time analytics pipelines** with Kafka, Spark Structured Streaming, EMR, Kinesis Data Streams, and Lambda to power **fraud detection, order tracking, and inventory forecasting**, reducing data latency from 15 minutes to under 20 seconds across **15M+ daily events**.
- Migrated **200+ legacy Teradata ETL jobs** to PySpark, Airflow, and dbt, eliminating on-prem dependencies and achieving **annual cost savings of \$1.8M**.
- Developed interactive **executive dashboards and operational monitoring** using Apache Druid, Tableau, CloudWatch, and QuickSight, enhancing decision-making across merchandising, pricing, and logistics by **45%**.
- Implemented **CI/CD pipelines** with Terraform, Jenkins, GitHub Actions, and AWS CodePipeline, reducing deployment errors by **40%** and accelerating feature rollout across multiple teams.
- Enforced **end-to-end data governance and security** with IAM, KMS encryption, Lake Formation policies, CloudTrail, CloudWatch Logs, and row-level security, ensuring **PCI-DSS and SOX compliance**.
- Mentored a **team of 7 engineers**, establishing best practices for coding, pipeline development, data quality, and cloud deployment, increasing team productivity by **35%**.
- Partnered with stakeholders across data science, merchandising, and logistics to define **data strategy and architecture**, aligning engineering initiatives with business goals and generating **multi-million-dollar impact**.
- Developed robust **ETL pipelines using PySpark** to clean, transform, and enrich data across various data sources (MySQL, NoSQL databases, Snowflake, MongoDB).
- Utilized PySpark's capabilities for data manipulation, aggregation, and filtering to optimize large-scale data preparation and reduce transformation time.
- Built and maintained **Kafka producers and consumers** to handle real-time data flow between applications, ensuring reliable and efficient data exchange.
- Employed **AWS Redshift** for efficient storage and analytical querying of terabytes of structured and semi-structured retail data.
- Loaded structured and semi-structured data from **MySQL tables** into Spark clusters using **Spark SQL and DataFrames API**.
- Designed, built, and managed complex **data pipelines using Apache Airflow**, orchestrating data workflows across multiple systems and ensuring timely execution of ETL processes.

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- Automated **end-to-end workflows** by scheduling and monitoring tasks with Airflow DAGs (Directed Acyclic Graphs), improving efficiency and reliability of big data processing pipelines.
- Designed and implemented **data ingestion pipelines using AWS Lambda** to efficiently bring data from diverse retail systems into the AWS S3 data lake.
- Constructed **AWS Managed Streaming for Kafka (MSK)** pipelines to deliver data streams from Target's APIs to processing endpoints such as Spark clusters in Databricks, Redshift, and Lambda.
- Ingested large-scale data streams from company **REST APIs into EMR clusters via AWS Kinesis**, enabling real-time analytics on transactional and customer behavior data.
- Processed streaming data from **Kafka brokers using Spark Streaming** for real-time analytics, applying explode transformations for data expansion.
- Developed complex **HiveQL queries** for data extraction, transformation, and reporting, enabling detailed sales, inventory, and customer insights.
- Utilized **Python and SQL** for large-scale data manipulation by joining datasets and extracting actionable insights from diverse retail data sources.
- Deployed, configured, and managed **Kafka clusters** to ensure high availability, scalability, and performance of data streaming and messaging systems.

Environment: AWS (S3, Glue, Redshift, Lake Formation, Athena, EMR, Kinesis, Lambda, MSK, CloudWatch, CloudTrail, CodePipeline), PySpark, Apache Spark, Kafka, Airflow, dbt, Terraform, Jenkins, GitHub Actions, Python, SQL, HiveQL, Databricks, Snowflake, MongoDB, MySQL, Apache Druid, Tableau, QuickSight, Data Lakehouse Architecture, Real-time Data Streaming, ETL/ELT Development, Data Governance, IAM, KMS Encryption, Row-level Security, CI/CD Automation, Big Data Processing, Cloud Data Migration, Data Quality & Validation, Performance Optimization, and Agile/Scrum Methodology.

Senior Data Engineer (Azure)

HSBC, New York, NY, Sep'21 – Oct'23

Summary: Led the design and implementation of cloud-based data platforms on Azure to support **HSBC's financial analytics ecosystem**. Built scalable ETL pipelines, enabled advanced reporting for trading, risk management, and portfolio analytics, and delivered enterprise-grade dashboards driving data-informed decisions across investment, compliance, and operational functions. Oversaw data quality, validation, orchestration, CI/CD, and mentorship of data engineering teams.

- Built a **cloud-native data platform on Azure** using **Data Lake Gen2, Data Factory, Synapse Analytics, Databricks, Azure SQL, Event Hub, and Key Vault**, consolidating 10TB+ of structured and semi-structured retail, sales, and supply chain data.
- Developed **automated ETL pipelines** with **PySpark, Airflow, and dbt**, integrating validation, testing frameworks, and lineage tracking to ensure accurate, reliable data delivery across departments.
- Designed and deployed **interactive dashboards** with **Power BI and Tableau**, providing real-time insights into **inventory levels, store performance, and customer purchasing patterns**.
- Implemented **CI/CD pipelines** using **Azure DevOps, GitHub Actions, and Terraform**, automating testing, versioning, and environment promotion — reducing release cycles by **50%** and improving reliability.
- Migrated **legacy SQL Server systems** to **Snowflake on Azure**, achieving a **3x performance boost** and cutting reporting latency from days to hours.
- Established **robust data governance and security** leveraging **Azure Key Vault, RBAC, encryption, masking, and auditing**, ensuring compliance with **PCI DSS** and corporate data protection standards.
- Mentored a **team of 6 engineers**, promoting best practices in **ETL development, CI/CD automation, and data quality assurance**, leading to measurable gains in delivery speed and system stability.
- Partnered with **business and data science teams** to develop **ML-ready data pipelines**, supporting initiatives like **demand forecasting, personalized recommendations, and store optimization**.
- Shifted orchestration to **Azure Data Factory**, enhancing data movement efficiency and scheduling flexibility.
- Ensured **secure data migration** by integrating **Azure Active Directory and Key Vault** for identity, access, and encryption control.

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- Optimized data storage architecture using **Azure Blob Storage, Data Lake Storage, and Synapse SQL Pools**, based on use-case requirements.
- Modeled **Hive partitions** and optimized data segmentation for faster query performance in **Azure HDInsight**.
- Executed **data migrations** from **Oracle and SQL Server** to **Azure HDInsight Hive and Azure Blob Storage** using **Azure Data Factory**, ensuring minimal downtime.
- Integrated **Azure Stream Analytics** for **real-time transaction and supply chain monitoring**, ensuring uninterrupted data flow.
- Managed migration operations via **Azure Monitor** and **Logic Apps**, automating alerts and performance tracking.
- Created **data frames** from diverse data sources (structured files, JSON, databases) within **Azure Databricks** for unified data analysis.
- Processed terabyte-scale datasets using **Spark (Scala/PySpark)** with efficient imports from **Azure Blob Storage**.
- Delivered **data visualization solutions** in **Power BI and Tableau**, converting raw data into actionable insights for **merchandising, marketing, and operations teams**.
- Authored **modular Python and Scala code** leveraging Azure Databricks libraries for ETL, analytics, and data enrichment.
- Automated ETL workflows with **UNIX shell scripts**, managing job scheduling, error handling, and file transfers through **Azure Blob Storage**.
- Administered **UNIX-based file systems and jobs** on **Azure Linux Virtual Machines** to ensure reliability and process continuity.

Environment: Azure Data Lake Gen2, Azure Data Factory, Azure Synapse Analytics, Azure Databricks, Azure SQL Database, Azure Event Hub, Azure Key Vault, Azure Active Directory, Azure HDInsight, Azure Blob Storage, Azure Stream Analytics, Azure Monitor, Logic Apps, Snowflake, dbt, Apache Airflow, PySpark, Spark (Scala/PySpark), Hive, Netezza, SQL Server, Oracle, Power BI, Tableau, Terraform, GitHub Actions, Azure DevOps, CI/CD, Data Modeling, ETL/ELT, Data Pipeline Automation, Data Quality & Validation, Data Governance, RBAC, Encryption & Masking, UNIX Shell Scripting, Python, Scala, JSON, Data Migration, Real-Time Data Streaming, Data Visualization, Machine Learning Data Preparation, Event-Driven Architecture, Metadata Management, Cloud Security, Performance Optimization.

Sr. Data Engineer

Dominion Energy, Richmond VA, Jan'19 – Aug'21

Summary: Led the modernization of Dominion Energy's data ecosystem through a **hybrid AWS-Azure architecture, building real-time grid performance** monitoring, predictive analytics, and enterprise data platforms to enhance operational reliability and compliance. Designed and deployed scalable ETL, streaming, and reporting solutions supporting smart grid, asset management, and regulatory reporting functions.

- Designed **event-driven pipelines** with **Kinesis, Step Functions, and Lambda** to process smart meter, IoT sensor, and SCADA data with **sub-second latency** for outage and demand forecasting.
- Migrated legacy **regulatory and compliance reporting infrastructure** to **Snowflake**, optimizing energy distribution analytics and operational reporting with **Spark** and **fact-dimension schema models**, reducing query execution time by **40%**.
- Automated **ETL workflows in AWS Glue**, integrating grid telemetry, customer usage, and asset management data into **Redshift/Snowflake** for downstream analytics and reliability studies.
- Delivered **real-time operational dashboards** on grid stability, load forecasting, and energy consumption using **Kinesis Data Analytics, Glue, Redshift, DynamoDB, and EMR**.
- Orchestrated pipelines with **Apache Airflow**, streamlining **end-to-end data workflows** across transmission, generation, and customer service analytics.
- Designed and orchestrated **ETL pipelines in Azure Data Factory**, integrating data from **smart grids, asset systems, and ERP platforms** into **Azure Data Lake (ADLS)** for enterprise-wide analytics.

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- Built **data marts in Azure Synapse Analytics**, enabling faster **regulatory reporting, demand analytics, and financial forecasting** with curated and auditable datasets.
- Delivered **self-service Power BI dashboards** for operations, maintenance, and finance teams, improving visibility into **asset health, energy output, and cost optimization**.
- Optimized **pipeline monitoring** with **Azure Monitor** and **Log Analytics**, improving system uptime and reducing downtime for mission-critical **energy infrastructure workloads**.
- Migrated **legacy SQL Server systems** into **Azure SQL Database**, modernizing analytics infrastructure for **asset performance, grid optimization, and energy trading**.

Environment: AWS (Kinesis, Lambda, Glue, Redshift, DynamoDB, EMR, EC2, RDS, VPC, Route 53, IAM), Azure (Data Factory, Synapse Analytics, Data Lake, Key Vault, Monitor, Log Analytics, Active Directory), Snowflake, Databricks, Apache Spark, PySpark, Apache Airflow, Step Functions, Power BI, SQL, Python, SCADA Data Integration, IoT Data Processing, Event-driven Architecture, ETL/ELT Pipelines, Data Lakehouse Architecture, Data Governance, NERC CIP Compliance, SOC 2, ISO 27001, and Performance Optimization.

Big Data Engineer

Chevrolet, Detroit MI, Sep'16 – Dec'18

Summary: Led the modernization of Chevrolet's analytics platform by designing real-time data ingestion and processing pipelines across Azure and Databricks. Delivered sub-minute transaction insights, enabling real-time customer analytics, predictive maintenance, and enhanced retail operations.

- Designed and implemented a real-time customer transaction and inventory data processing platform using **Kafka** and **Azure Event Hubs**, reducing data delivery delays from hours to under a minute.
- Enabled ingestion from **IoT-enabled retail devices** (POS systems, smart fitting rooms, and in-store sensors) via **Azure IoT Hub**, powering real-time store operations and customer experience workflows.
- Developed **metadata-driven ingestion pipelines** using YAML/JSON configurations, reducing manual coding and accelerating feature delivery.
- Automated ingestion of customer, sales, and merchandising data from structured and semi-structured sources using **Azure Data Factory (ADF)** and **SQL Server**.
- Optimized analytics performance using **Azure Synapse Analytics**, improving dashboard response times for sales trends and loyalty insights.
- Built modular **ETL pipelines** with **ADF, Delta Lake**, and **Databricks (PySpark)** for e-commerce and inventory data processing.
- Replaced fragile legacy scripts with **parameterized ETL templates**, improving workflow reusability and reliability.
- Enforced **data governance** on **ADLS Gen2** through **Azure Purview** and **RBAC**, ensuring compliance and secure data access.
- Deployed **Terraform** for Infrastructure-as-Code (IaC), ensuring environment consistency and faster provisioning.
- Implemented **CI/CD workflows in Azure DevOps**, enforcing schema validation and version control.
- Established monitoring using **Azure Monitor**, **Log Analytics**, and **Application Insights**, proactively resolving ETL performance issues.
- Implemented **Change Data Capture (CDC)** and incremental loading via **ADF Mapping Data Flows** to optimize data refresh cycles.
- Collaborated with **business intelligence, data science, and compliance** teams to align architecture with analytics and regulatory standards.
- Conducted internal workshops on **ADF tuning, Databricks optimization, and data security best practices**.

Environment: Azure Data Factory, Databricks (PySpark), Delta Lake, Azure Synapse Analytics, Azure Event Hubs, Kafka, Azure IoT Hub, Azure Purview, Azure Key Vault, Terraform, Azure DevOps, SQL Server, Azure Monitor, Log Analytics, Application Insights, CI/CD, ETL, YAML/JSON, CDC, Data Governance, RBAC, PCI-DSS

Data Engineer

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Pfizer., New York NY, Jan'15 – Aug'16

Summary: Led the design and implementation of on-premises big data pipelines and real-time analytics solutions on the **Hadoop** ecosystem to support Pfizer's R&D, manufacturing, and predictive maintenance programs. Delivered scalable, secure, and ML-ready data infrastructure, improving operational visibility and enabling advanced analytics across clinical and production environments.

- Built end-to-end ETL pipelines using Apache Sqoop, Flume, Spark, and Hive on a multi-node Cloudera Hadoop distribution (CDH 5.x), processing terabytes of structured and semi-structured data daily from laboratory, MES, and ERP systems.
- Implemented near-real-time streaming with Kafka, Spark Streaming, and HBase, enabling sub-minute anomaly detection in sensor and equipment telemetry across manufacturing plants.
- Developed data quality and validation frameworks in Oozie and Python, enforcing schema consistency and record-level checks across ingestion workflows.
- Created analytical data marts in Hive and Impala, serving R&D and production dashboards in Tableau and QlikView with daily refreshed KPIs and quality metrics.
- Automated batch workflows with Apache Oozie and Azkaban, integrating with Git and Jenkins for version-controlled deployments and job scheduling.
- Partnered with data scientists to provision ML-ready datasets in Parquet and Avro formats, supporting predictive maintenance models using Spark MLLib and Python scikit-learn.
- Integrated legacy Oracle and SQL Server systems using Sqoop and custom Python connectors, ensuring secure and incremental CDC extraction into HDFS.
- Engineered large-scale data transformations with Spark and Pig, optimizing MapReduce jobs for performance, compression, and partitioning efficiency.
- Implemented centralized logging and monitoring via Cloudera Manager, Nagios, and Ganglia, providing visibility into job health, cluster utilization, and SLA compliance.
- Designed data lake governance policies with HDFS directory permissions, Kerberos authentication, and Apache Ranger, aligning with FDA 21 CFR Part 11 and HIPAA requirements.
- Deployed HBase tables for low-latency operational lookups and integrated them with Spark for hybrid analytical workloads.
- Created metadata-driven ingestion frameworks using XML/JSON configs to standardize ETL logic across R&D and manufacturing domains.
- Optimized storage and compute costs by implementing YARN resource tuning, data compression (Snappy, LZO), and tiered storage between HDFS and NFS archival.
- Coordinated cross-functional collaboration between infrastructure, QA, and analytics teams to migrate legacy ETL from Informatica to Spark-based pipelines, reducing processing time by over 50%.

Environment: GCP (BigQuery, Dataflow, Pub/Sub, Composer, Dataproc, Cloud Storage, Cloud Build), Apache Airflow, Apache Spark, Python, Kafka, Terraform, GitHub Actions, DBT, Oracle, Hive, Snowflake, Looker, Tableau, Data Studio, IAM, VPC Service Controls, Cloud Monitoring, Dataprep, CI/CD, ETL, Data Modeling, Data Governance, Predictive Analytics

Data Analyst

State Farm Insurance, Bloomington IL, Jan'14– Dec'14

- Collected, cleaned, and analyzed large datasets from policy, claims, and customer systems to identify trends and improve underwriting accuracy.
- Developed interactive dashboards and performance reports using Tableau and Excel, enabling data-driven decision-making for business stakeholders.
- Conducted statistical analysis and predictive modeling using SQL and Python to forecast claims frequency and customer retention.
- Automated routine data extraction and validation processes, reducing manual reporting time by 30%.
- Collaborated with cross-functional teams to define key performance indicators (KPIs) and improve data quality and governance across business units.

Education:

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Bachelor of Science - Data Science

Michigan State University, East Lansing, MI

Certification:

Software Engineering Program, Certificate

Springboard, San Francisco, CA